

Spatial audio technology for telepresence

M P Hollier, A N Rimell and D Burraston

As people start to exploit new telepresence technologies to meet and work, they will be able to exploit all of their senses as they transmit, receive, and monitor information. An essential part of such three-dimensional spaces is the audio landscape. People are able to detect a wide variety of sounds and separate them in space. Spatial separation improves the detection and intelligibility of speech from multiple talkers, and enables simultaneous monitoring of multiple information streams through the use of multiple alert sounds. On-going research at BT Laboratories into spatial audio has resulted in a number of leading edge demonstrations and patent applications. This paper introduces the technologies employed to create spatial audio for real-time synthetic worlds including single and multiple users, and non-ideal acoustic environments.

1. Introduction

A key requirement to enhance the usability of a range of telepresence applications is the exploitation of all the sensory modalities, thus maximising naturalness. Spatial audio has a vital role to play in enhancing speech communication performance and naturalness in synthetic spaces by giving directionality to individual sound sources and providing feedback about the virtual space in which the user is located. When groups of people meet in synthetic spaces either for business or recreation it is possible to provide services and features which are not available in a face-to-face meeting. In particular, interaction with intelligent agents, manipulation of virtual 3-D objects, data sharing, and automatic note taking may all seem quite natural in the enhanced meeting spaces of the future.

1.1 Audio landscape

The sensory modality which dominates perception is normally assumed to be vision and this has been reflected in the attention paid to the visual components of virtual world developments to date. However, for communication it can be readily argued that hearing is the most indispensable modality — imagine a videoconference with no audio! More generally, hearing may provide very compelling cues which offer powerful support for the naturalness of a synthetic environment. For example, navigation cues in a synthetic space are particularly important to avoid disorientation and to facilitate efficient location of persons or items of interest. The required cues can be provided by associating an audio landscape with the world; the location of other talkers and objects of interest in close proximity can then be readily identified by the direction and character

of their audio signal (compare this with the poor usability of an auxiliary visual display).

1.2 3-D spatial sound

The work reported here concerns high-quality audio systems capable of reproducing a virtual audio source for either a 2-D (full 360 ° plane around the user), or 3-D (any direction in three dimensions) space. This approach must be distinguished from other pseudo-3-D audio and surround sound systems found in existing distributed virtual worlds and home cinema systems. For example, a typical home cinema surround sound system such as Dolby ProLogic does not usually produce 2-D spatial sound (a rear image can be created using advanced steering techniques [1]). DTS and Dolby-AC3 systems provide 5.1 discrete channels (5 full bandwidth channels and a sub-woofer) and therefore could more readily produce a full range of virtual source positions if source mastering and loudspeaker placement for replay were appropriate. In general, of course, it is not required, or desirable, to place distinct virtual sources behind the listener for home cinema since there would be nothing there to see if the user were to turn around and look. However, in a synthetic environment it is entirely possible that an object of interest might be located behind the user who would be able to ‘turn around’ and look at it within the virtual space.

Research into the exploitation of spatial sound at BT Laboratories (BTL) has been conducted within a team responsible for more general research into perception and the modelling of the human senses for performance

assessment. This knowledge may be exploited to design attention grabbing alarms and to avoid startling or disorienting the individual.

Several of the key technologies for practical spatial audio reproduction have been known for many years [2]. However, their integration into real-time interactive systems has required many advances to be made particularly in system architecture and control. BT has worked with a number of collaborative partners to develop the new techniques reported and relevant partners are acknowledged at relevant points in the paper. This paper discusses the technologies used to create perceptually compelling spatial audio for synthetic worlds.

1.3 *Unification of visual and audio worlds*

The bias towards the visual aspects of virtual environments is particularly evident in the files and structures used to define most virtual world software products. The emphasis in the geometry definition, motion description, and information stored about the physical properties of the world's structural materials are all oriented towards the requirements of video rendering. This is quite reasonable as a starting point since numerous developments in graphical handling and processing were required to produce the visual virtual environments which are currently available. Unfortunately the geometric and materials definitions which have evolved for visual virtual worlds are not always ideal for an acoustic model which requires different geometric resolution and additional information about the structure of the world.

Models of virtual acoustic environments do exist and find practical application. In particular, a number of modelling packages are available for auralising buildings such as concert halls while these are still in the planning stage. Given the capital value of such projects it is a testament to modern auralisation packages that they are deemed sufficiently accurate for this kind of application.

To progress the application of advanced spatial audio techniques within unified audiovisual worlds, a partial integration of existing visual and audio models has been developed at BT Laboratories (BTL). In one system the visual world software was modified to extract the spatial co-ordinates of listener and source positions. This information is then coded as TCP/IP packets and transmitted to an audio server which runs a separate world model with the required acoustic properties. The audio server then spatialises the audio according to the geometry information received and introduces the appropriate room acoustics such as reflections and reverberation according to the room model loaded. It is possible to move from room to room within the virtual world which allows the different room models to exhibit widely varying characteristics, such as a dead recording studio through to a large reverberent church. The

unification of the visual and audio parts of virtual worlds remains the subject of on-going research.

2. **Spatial audio requirements**

The emphasis of the spatial audio research reported here has been on high-quality fully spatialised systems. Lower performance/surround sound systems are extensively discussed elsewhere and are, in general, unable to position a virtual sound source at all points around the user.

Spatial audio requirements can be categorised by the number of users, restrictions on sound reproduction techniques, and the type of spatial audio percept required. Some examples are listed in the following sections.

2.1 *Single users with headphones*

A common situation is expected to be a single user in front of a VDU, who is likely to look continually at the screen. In this case the headphones need not be motion tracked since the user will face forwards. An advantage of headphones is that they do not typically produce sound which is intrusive to others. When one or more users are equipped with headphones and have freedom to move about, head-tracking is necessary to keep the audio landscape static; for example as a user turns round a sound source should appear to stay still, coming from behind the user when he turns his back towards it.

2.2 *Single users with loudspeakers*

A typical configuration is for the user to have a pair of loudspeakers either side of a VDU. Transaural reproduction has the advantage that it requires two conventionally located loudspeakers and can provide very wide angle frontal spatialisation. However, it cannot currently provide compelling audio images behind the user.

A development of this technology, known as Soundbubble, with improved rear spatial images is under development between BT Laboratories and the University of Essex.

2.3 *Multiple users*

When multiple users require a consistent spatial audio percept, conventional stereo, transaural and ambisonic all have limitations due their limited sweetspot. The sweetspot is the ideal listening position, away from which spatial sound reproduction is impaired. Various techniques have been investigated to provide spatial sound to groups of users. Steered mono offers some advantages for video-conferencing since it provides spatialisation which is independent of listener position, can be achieved with a single audio channel and simple control information, and

results in an echo-cancellation requirement that is potentially simpler than multi-channel techniques. True 3-D spatial sound implies that all users will hear a virtual sound source constantly following any desired trajectory. A technique developed at BT to provide spatial sound for a group of users in the VisionDome immersive environment is described in the body of the paper.

3. Spatial audio technologies

A variety of spatial audio techniques have been investigated and developed at BTL to suit a range of applications [3]. A number of these technologies are introduced below.

3.1 Binaural

A binaural signal (recorded with a dummy head or synthesised) recreates the soundfield present at the entrance to the ear canal when listening to an actual sound source [4]. The sound waves impinge the outer ear (pinna) at a given angle, the pinna then filters the sound with an angle-dependent transfer function (head-related transfer function — HRTF). The combination of HRTFs and time delay enable the listener to locate sounds — HRTFs have the greatest influence at mid and high frequencies, and time delay at low frequencies. Figure 1 illustrates the encoding process where HRTFs are encoded into each ear's signal. To prevent the sound being filtered by the pinna twice it is necessary to use headphones to listen to the binaural signal. A dummy head recording is made with an anatomically accurate artificial head with microphones positioned at the entrance to the ear canal whereby the artificial pinnae provide the HRTF filtering. Synthesised binaural encoding requires the signal to each ear to be convolved with the appropriate HRTF — for a moving virtual source location a considerable amount of convolution must take place. The HRTFs are measured with a dummy head or by placing microphones inside a subject's ear canal with the system impulse response being recorded under anechoic conditions.

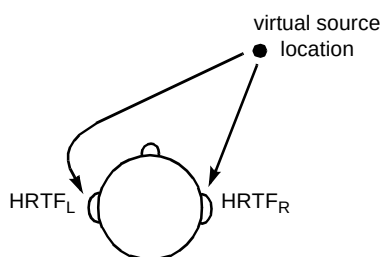


Fig 1 Binaural signal encoding.

Binaural encoding produces very realistic sounding audio especially when it also encodes the HRTF of each individual reflection. All of the processing is carried out at the encoding stage and the spatialised audio can be reproduced on conventional audio equipment. The principal disadvantage is the necessary use of headphones.

3.2 Headtracked binaural

Because binaural signals are heard with headphones, any movement of the head will result in the same movement of the spatialised audio signal. In telepresence systems it may be desirable to keep the sound location static with respect to the physical environment, resulting in movement as the listener rotates their head. By applying different HRTFs for different head positions the sound source location can be kept static; a typical practical implementation [5] uses 128 different HRTFs.

The solution comprises a pair of headphones with a rotational position sensor mounted on top and DSP-based hardware which encodes each of the two audio signals with the appropriate HRTF filters. Figure 2 shows the effect of head movement both with and without head tracking applied.

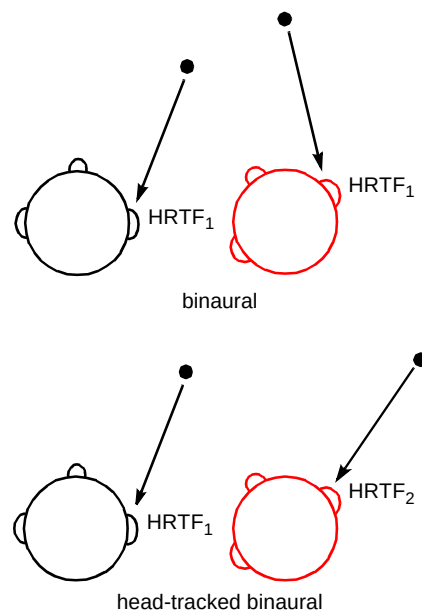


Fig 2 Binaural signal with head-tracking

Unfortunately, head-tracking requires real-time digital filtering; however, this is offset by audio image locations remaining static during listener head movement — this is useful in virtual meeting environments as each participant's voice has a constant localisation with respect to the video presentation. Like standard binaural systems, headphones are required for sound reproduction; however, they now require a positional sensor to be attached or built in.

3.3 Transaural

A disadvantage of binaural encoding is the necessity of headphones for playback. Transaural systems use crosstalk cancellation to enable HRTF processing to be used with loudspeakers. Any number of loudspeakers may be used and they may be located at any angle with respect to the listener. Figure 3 describes the principles behind the shuffler crosstalk cancellation network introduced by

Cooper and Bauck [6]. The signal S represents the transfer function from the loudspeaker to the nearest ear, and the signal A represents the transfer function from the loudspeaker to the opposite ear. The functions S and A are defined as the HRTFs from each speaker to each ear. In this example, both loudspeakers are at an equal angle from the listener resulting in the same values of A and S for left-to-right and right-to-left signal paths. Where the loudspeaker arrangement is not symmetrical there will be two values of both S and A , and if there are N loudspeakers there will be N values of S and A .

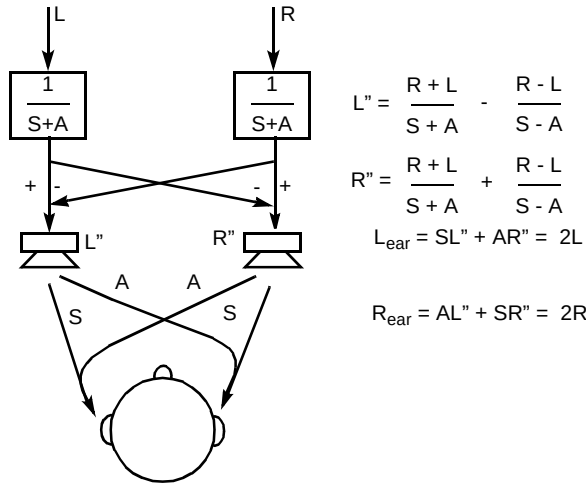


Fig 3 Crosstalk cancellation.

Transaural coding gives good auditory localisation and avoids the use of headphones in the playback system. Transaural is more suitable for individual workspace implementation rather than a multi-user environment as the correct combination of signals from the N speakers only occurs within a relatively small sweetspot.

3.4 Ambisonics

Ambisonic theory presents a solution to the problem of encoding directional information into an audio signal [7 —

9]. The signal is intended to be replayed over an array of at least four (for a horizontal plane only — pantophonic system) or eight (for a 3-D space — periphonic system) loudspeakers. Like a binaural system, the sound space can be recorded with a specifically designed microphone [10] or synthesised.

The signal, termed B-Format, consists of three components for pantophonic systems (W the ambient component, X the front-back component and Y the left-right component) and four components for periphonic systems (W , X , Y and Z the up-down component).

Figure 4 shows the geometry used in the B-Format encoder, the equations shown define the W , X , Y and Z components of the signal, where the 3-D spatialised sound to be encoded is placed within a unit sphere.

During the encoding stage no knowledge of the loudspeaker positioning is necessary — the decoder is programmed with the loudspeaker lay-out. The output at each of N speakers in an equally spaced pantophonic array is given by:

$$P_N = \frac{1}{N}W + 2X\cos(\phi) + 2Y\sin(\phi)$$

where W , X and Y are defined as:

$$W = S \cdot \frac{1}{\sqrt{2}} \quad X = S \cdot \cos(\phi) \quad Y = S \cdot \sin(\phi)$$

The equation can be expanded for a periphonic array with a regular eight speaker array or an irregular N speaker array. A regular array gives:

$$P_1 = \frac{1}{4}W + \frac{2X}{\sqrt{2}} + \frac{2Y}{\sqrt{2}} \quad P_2 = \frac{1}{4}W - \frac{2X}{\sqrt{2}} + \frac{2Y}{\sqrt{2}}$$

$$P_3 = \frac{1}{4}W + \frac{2X}{\sqrt{2}} - \frac{2Y}{\sqrt{2}} \quad P_4 = \frac{1}{4}W - \frac{2X}{\sqrt{2}} - \frac{2Y}{\sqrt{2}}$$

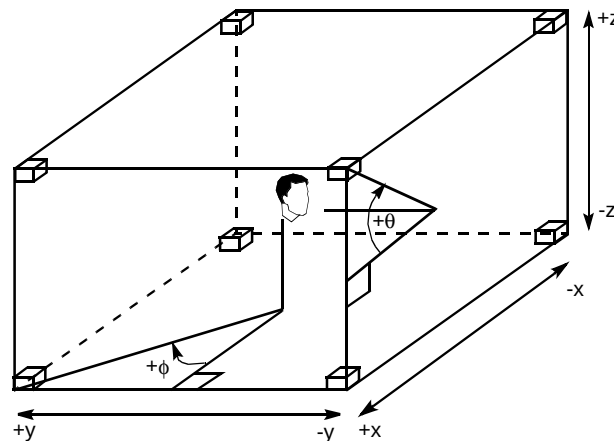


Fig 4 Ambisonic system geometry.

$$W = S \cdot \frac{1}{\sqrt{2}}$$

$$X = S \cdot \cos(\phi) \cdot \cos(\theta)$$

$$Y = S \cdot \sin(\phi) \cdot \cos(\theta)$$

$$Z = S \cdot \sin(\theta)$$

Ambisonic encoding provides a realistic spatial audio percept; however, because the theory is based on wavefront reconstruction the correct summation of the loudspeaker outputs only occurs within a small sweetspot, as with transaural systems, making an ambisonic system most suitable for personal workspace environments.

While one advantage of ambisonic systems over transaural systems is that the speaker lay-out does not need to be known at the encoding stage, a greater number of loudspeakers are required.

3.5 Strategies for multiple users

Nonlinear panning

In multi-user environments (such as the ARC VisionDome, described below in section 4.3), it is necessary to provide spatialised audio for a group of users. The two loudspeaker-based methods described above only fully spatialise the audio within a small sweetspot area. The use of intensity panning across the N loudspeakers gives listeners who are situated near a loudspeaker an improved spatial percept, due to the elimination on the ambisonic ambient sound signal (W). However, the signal will now consist of N loudspeaker position-dependent channels rather than the 2 or 3 of a B-Format signal.

Where the spatialised audio is synchronised to a moving video image projected on a concave screen (as in the ARC VisionDome — a typical example of a multi user environment), a non-linear panning is required to keep the perceived sound and video image locations synchronised.

B-format warping

The aim of the warping algorithm is to change from a linear range of x and y values to a nonlinear range. Consider the example where a sound is moving from right to left; in order to remain consistent with the visual display on a large concave screen, the sound needs to move quickly at first then slowly across the centre and finally quickly across the far left-hand to provide a corrected percept.

B-Format warping takes an ambisonic B-Format recording and corrects for the perceived nonlinear trajectory. The input to the system is the B-Format recording and the output is a warped B-Format recording (B'-Format recording). The B'-Format recording can be decoded with any B-Format decoder allowing the use of existing decoders. An ambisonic system produces a sweet-spot in the reproduction area where the soundfield reconstructs correctly and in other areas the listeners will not experience correctly localised sound.

Given the B-Format signal components, it is possible to determine the original values of x and y as the original signal can be reconstructed to give $S' = W\sqrt{2}$ from which values of x and y can be determined:

$$x' = \frac{X}{S'} \quad \text{and} \quad y' = \frac{Y}{S'}$$

so

$$x' = \frac{X}{W\sqrt{2}} \quad \text{and} \quad y' = \frac{Y}{W\sqrt{2}}$$

A suitable warping method is obtained by squaring the value of x and y , or in the more general case by raising it to a power, $2i$:

$$X' = X \cdot (x')^{2i} \quad \text{and} \quad Y' = Y \cdot (y')^{2i}$$

$$X' = X \cdot \left(\frac{X}{W\sqrt{2}}\right)^{2i} \quad \text{and} \quad Y' = Y \cdot \left(\frac{Y}{W\sqrt{2}}\right)^{2i}$$

$i = 0$ no warping

$i > 0$ nonlinear warping

Because the values of x' and y' may be positive or negative it is necessary to ensure that X' and Y' have the same sign as X and Y respectively. This is achieved by making the multiplier an even number, i.e. by raising

$$\left(\frac{X}{W\sqrt{2}}\right)$$

and

$$\left(\frac{Y}{W\sqrt{2}}\right)$$

to an even power ($2i$).

The resultant signal, X' , Y' and W is referred to as the B'-Format signal, as shown in Fig 5.

A further adaptation of the B-Format signal to reduce or remove anti-phase components has also been investigated to avoid misperception of spatial sound by listeners close to ear channel speakers.

Aspects of the manipulation of the B-Format signal developed at BTL are the subject of a patent application. Warping the B-Format signal has the advantage that standard recordings can be manipulated to suit specialist environments such as the VisionDome.

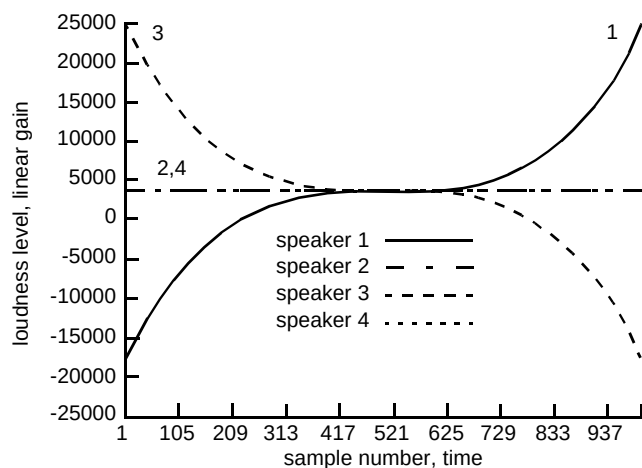


Fig 5 Nonlinear loudspeaker feed profiles after transformation of the B-Format signal to suit a non-ideal listening room.

4. Experimental spatial audio systems at BTL

This section of the paper introduces a series of demonstrators which were developed to prove out a series of spatial audio concepts:

- Cube World represented an early realisation of a real-time interactive 3-D world with high-quality spatial sound,
- BIER-link introduced spatial audio to a distributed shared space for two people,
- spatial audio to suit groups of users and non-ideal acoustic environments were implemented for the BT VisionDome,
- RevRooms introduced the concept of having different acoustics in a number of separate rooms,
- NSAS demonstrated the benefit of spatial sound in multi-user distributed environments.

4.1 Cube World

The first steps in creating a virtual environment with spatial audio was realised with Cube World. This was an elementary visual representation, consisting of a large room with four cubes inside. The cubes were positioned at different heights towards the four corners of the room. Each cube has a different sound source that emanates from its location in 3-D space. The sound sources used were ambulance, lorry, plane and car horn. This was a fully interactive, real-time, multiple sound source, audio/video environment. A mouse or trackball was used to navigate around the world. The graphics were displayed on to a large back projection video screen. An eight-speaker array was used for the spatial audio playback.

The system consisted of an SGI Indigo 2 workstation, a Lake Huron Digital Audio Workstation and an eight-speaker array. The virtual world was rendered on the Indigo 2 and the control information for user and sound source position were sent to the Huron via TCP/IP sockets in real

time. The Huron rendered a B-Format ambisonic soundfield, with Doppler shift, for the eight speakers in real time. This is continuously updated as the user navigates around the world, giving a perceptually compelling 3-D soundfield.

4.2 BIER-link

The BIER (Broadband Interactive Enhanced Reality) link was developed from the body tracking used in the ALIVE technology from the MIT Media Lab [11]. The BIER link enables two people to have a virtual conference over ISDN with a multi-speaker ambisonic system. The audio was encoded using a G.722 CODEC giving a 7 kHz bandwidth and transmitted with control information over ISDN. At each end the user is free to move around in an area 4.5 m × 4.5 m, their position being monitored by a video camera. This area is the shared space that both users inhabit for the duration of the conference. The body tracking software allows for rudimentary gesture input to animate the user's avatar. The position of the user's avatar is continually adjusted in real time to match the user's real location. Figure 6 shows the virtual environment inhabited by two users.

With the location of the users known, it is possible to transmit the local audio signal to the remote user and position this in 3-D space. Localisation problems arise when the user moves away from the geometric centre of the sound system and hence out of the sweetspot. A novel means to overcome this problem for a single user is to transmit the position co-ordinates to the decoder in real time. These co-ordinates are then used to update the decoder to relocate the sweetspot to the user's current location. This approach, including the performance of a real-time system, is reported in Burraston et al [12].



Fig 6 Two users rendered in the BIER shared environment.

4.3 BT VisionDome

The BT VisionDome is an interactive, digital virtual-reality environment, which enables group interaction around common data or a shared application [13]. The use of a hard plastic dome as the listening room creates many acoustic problems, mainly caused by multiple reflections [14]. The problem was further compounded by the requirement to provide spatial audio for multiple listeners inside the dome. The multiple reflections were reduced by covering the inside of the dome in acoustic tiles.

Reproducing spatial audio for multiple listeners presented another new problem. In this case there is no sweet spot to reconstruct the audio wavefront, because all the listeners are spread out inside the dome. To overcome this problem, a nonlinear panning strategy was developed in collaboration with Lake DSP, and implemented on their HURON system [15]. The system receives the X, Y, Z location of the sound source, the X, Y, Z locations of all the speakers and the monophonic sound source. The sound is then panned from one position to another using a nonlinear panning characteristic. This system worked well in the dome.

The nonlinear panning technique described above can only be used where the co-ordinates are known for each sound source, e.g. in an interactive computer program. A BTL-developed B-Format warping algorithm allows pre-recorded, standard, B-Format material to be acoustically rendered in the VisionDome to provide compelling spatial sound for a group of listeners.

4.4 RevRooms

RevRooms (Reverb Rooms) was the first system developed that used spatial audio rendering coupled with room acoustic simulation, known as auralisation [16]. The spatial audio used a transaural playback system to render sound from two loudspeakers. The virtual world was a simple model incorporating four rooms of increasing size from small room to large hall. Each room had a fixed human shaped figure, with a small segment of looped speech playing from the co-ordinates of its head. The system ran in real time and allowed the user to freely travel around the environment. The hardware used to spatialise sounds and acoustically model room parameters was controlled using MIDI (Musical Instrument Digital Interface).

This system effectively demonstrated the ability to interact with a virtual world, while generating an auralisation in real time for each room. Users experienced a greater feeling of immersion in the environments due to the inclusion of room acoustics, which accurately matched the visual percept of room size. The ability to accurately model/synthesise room acoustics is essential in creating a compelling sense of presence. Figure 7 is a screen-shot of

the system, which shows the avatar of another user and a doorway to another room.

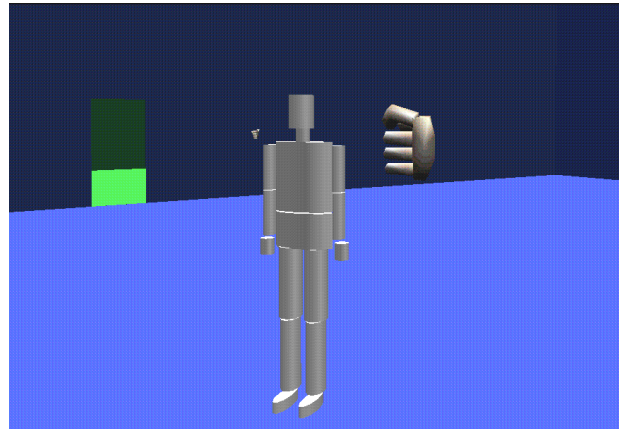


Fig 7 View of RevRooms environment.

4.5 NSAS (network spatial audio server)

NSAS is a current project which utilises several aspects of the technologies described above to provide a networked shared meeting space with high-quality spatial sound. Each participant receives a 3-D audio rendering of the soundfield around them in the virtual environment. This includes the virtual room acoustics of the environment and any sound source considered to be within a predetermined range. The sound sources include the speech of users in the environment and other sounds generated within the virtual environment. The virtual environment in the concept demonstrator consists of a number of rooms where users are allowed to enter and hold meetings. It would be feasible for groups of users to create conferences by providing lectures which would be audible to attendees within a designated room. The number of users is not fixed and individuals can enter or leave the environment at any time.

An overview of NSAS and its associated components is shown in Fig 8. Audio and control messages will be sent across a computer network. The system consists of three sub-systems and N number of clients. The three sub-systems are:

- network spatial audio server (NSAS) for rendering the spatial composite and auralisation for each client,
- audio services sub-system will provide the system with any data- or event-driven sounds, voice annotation/voice messaging and rudimentary data sonification,
- virtual world server will control the virtual environment for each user and the interface between the user and the other sub-systems.

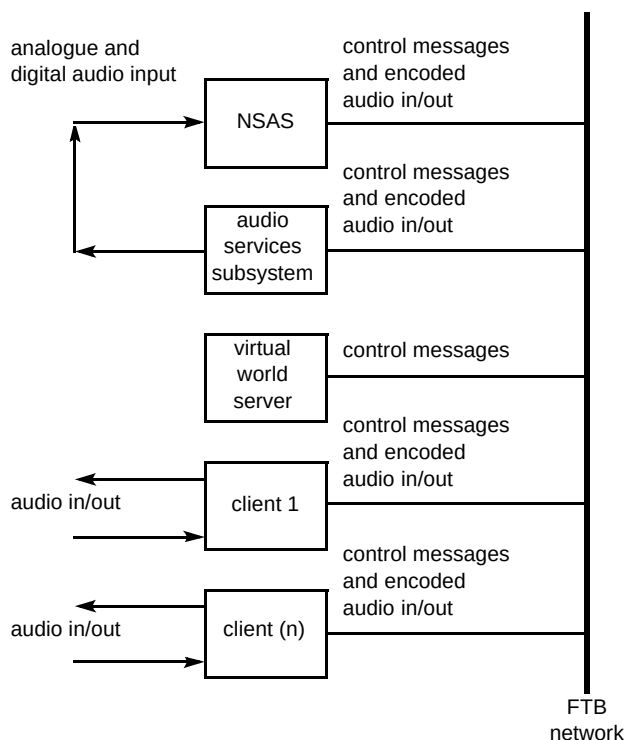


Fig 8 Overview of NSAS and associated components.

A prototype system exists as two clients, connected via balanced analogue cables, the NSAS server, the virtual world server and a small audio services sub-system. The virtual environment consists of a corridor with five doorways to enter any of the three meeting rooms or two lecture/conference rooms, each having simulated acoustics including doorway simulation. Each room contains a mannequin reciting a short monologue and one room also contains a moving sound source. This is useful for demonstrations, allowing the system to show its own capabilities.

When a user logs into the environment, their avatar is created and made visible to all other users. The avatar always represents the user's location within the virtual environment, and when they speak, their voice will always be at the same location as their avatar. The result is a natural, intuitive means to communicate within a virtual environment or conference.

5. Discussion and conclusions

Much of the leading edge development in virtual reality and multimedia has focused on the video component, with the audio part often being underdeveloped. BT's history as an audio services provider has given a valuable perspective for the development of advanced telepresence applications in which the importance of audio performance is acknowledged. The provision of high-quality spatial audio can greatly enhance the intelligibility of multiple voices and is an invaluable orientation and navigation aid

for a variety of applications including virtual spaces. Various spatial audio technologies have been summarised and a series of leading edge demonstrators which employ them have been introduced.

Acknowledgements

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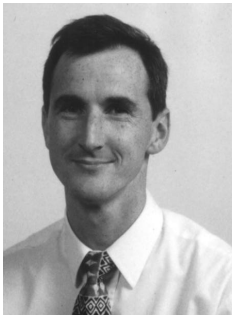
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Mike Hollier obtained a BEng (Hons) degree in Mechanical Engineering from Plymouth Polytechnic in 1987. After working as a hi-fi loudspeaker designer, he joined BT Laboratories to work on a variety of acoustics projects including the design of a novel noise-cancelling handset. He obtained an IOA diploma in acoustics in 1989, receiving an industry award for his research into the vibrational behaviour of light structures.

From 1990 his work included the research of objective methods capable of predicting the performance of nonlinear audio systems. He developed an analysis based on a model of human auditory perception, receiving his PhD from the University of Essex in 1995 together with the BT Postgraduate publication prize. Since January 1996 he has formed and led the Multi-Modal Perception group with projects on auditory, visual and multi-sensory perception. His team also carries out work on spatial audio for conferencing and virtual environments.

He is a visiting lecturer to the University of Essex, where he is a fellow of the Department of Electronic Systems Engineering, a Chartered Engineer, and a member of the IOA and AES.



Andrew Rimell completed a research MSc and a PhD at the University of Essex in 1993 and 1996 respectively. His PhD proposed a method of reducing the perceived crossover-induced errors in a multi-driver loudspeaker system through the use of a psycho-acoustically motivated error concealment algorithm. In 1996 he was awarded an IEE Premium Award for his work in modelling room acoustics.

He is now employed at BT Laboratories where he works in the Multi-Modal Perception group. His work includes research into spatialised audio for telepresence and the use of psychoacoustic models in the quality assessment of mobile telecommunications systems.

He is an active member of the IEE Professional Group Committee E-10 (Circuits & Systems), a member of the AES, and has presented numerous conference papers. His research interests include spatial audio, perceptual modelling, loudspeakers and signal processing.



Dave Burraston joined BT in 1981 as an apprentice, specialising in repeater station maintenance. After completing his apprenticeship he went on to work in various electronic workshops, testing and repairing electronic call-connect and computer systems at the component level. In 1989 he commenced full time study for a BSc in Computing Systems at Nottingham Trent University.

After graduation in 1993 he moved to BT Laboratories to join the Visualisation Research group, researching the technology and potential applications of data visualisation and virtual reality systems. He currently works in the Multi-Modal Perception group and is involved in research on spatial and non-spatial audio in shared virtual environments.